

Advanced Mathematical Tools for Management

This course aims to introduce or review the essential mathematical concepts, techniques, and tools necessary for understanding scientific literature and conducting research activities in various fields of quantitative management.

MATH 40604A, Fall 2026

[HEC Montréal](#)

Welcome to MATH 40604A!!

Welcome, and congratulations on starting a new term. My name is Khaled, I'm a PhD candidate in data science at HEC Montreal, and I will be teaching the English section of this class. The purpose of this course is to give you the mathematical and statistical background you need to feel comfortable in the rest of your program. We will review and build up on the following topics: modelling, linear algebra, functions, calculus, optimization, probability, and statistical inference. That is a wide territory to cover but you will find yourself familiar with some of the material. The course is new this year, and by the end of the term you should feel more comfortable around mathematical/statistical notation.

ZoneCours vs this website

You may have noticed that both sections are sharing the same ZoneCours which makes sharing files there a little bit complicated. This site will contain all the course material needed for the English section. However, I will still upload the same documents to ZoneCours for those who prefer it.

How the course works

Feel free to look through the [course outline](#) to see what each session covers. Because there is a big ground to cover, part of the reading will happen at home before class. Every week, I will upload two pre-class documents, one containing the notes and the other containing some exercises (with solutions provided). I have tried to make both as easy to follow as I can. If a topic gives you trouble, please skip it, make a note of it, and we will go over it together during class.

Before our first session

Our first class is Monday, August 24th at 8:30–11:30. I know that is early in the term, but I would be very grateful if you took a look at the Week 1 material [Week 1 - Modelling, sets, and algebra](#)

Until then, enjoy what is left of the break and get some rest. Feel free to email me if you have any questions. I am looking forward to meeting you all.

~ Khaled

Contact Information

- **Instructor:** Khaled Fouda
- **Office:** 4.455, Côte-Sainte-Catherine
- **Email:** khaled.fouda@hec.ca

Course details

- **Semester:** Fall 2026
- **Day:** Monday
- **Time:** 8:30-11:30
- **Room:** [Hélène-Desmarais, CSC](#)

Office hours

Office hours will be decided during the first class. Feel free to reach me anytime by email.

Weekly notes

Week	Topic	Notes
1	Modelling, sets, and algebra	Week 1

A note on the course documents: the notes are based on material developed by Fatiha Kacher, who teaches the French section. This site, and all material posted on it, is for our (English) section.